

Paper A v0.2 — The Physical (SU(3)) Yang–Mills Mass Gap via D_{IV}^5 Spectral Geometry. Companion to Paper B (substrate uniqueness). v0.2 UPGRADE (Tuesday 2026-06-23 + Wednesday 2026-06-24 AM): the §7 cross-channel glueball spectrum — NAMED-OPEN in v0.1 — has been DERIVED. The four ground channels follow a linear conformal-energy ladder off one verified number (the genus of $D_{IV}^5 = n_C$), landing the full J^{PC} spectrum in MeV at $<0.6\%$, with $2^{++} = g/n_C$ a BLIND leg (genus + spin only, lands because $g = n_C + \text{rank}$). The radial direction is settled LINEAR by Schur’s lemma (the Casimir is constant on an irrep), retiring the $m\text{-vs-}m^2$ ambiguity and yielding a clean negative on the spin-linear/radial-quadratic factorization. The glueball MULTIPLICITY is explained: $F \otimes F$ decomposes into the four irreducible J^{PC} operator channels {energy 0^{++} , topology 0^{-+} , curvature 2^{++} , derivative 1^{+-} }, each forced to exist by operator-algebra closure (the WHY, not just the WHAT). The experimental scalar-glueball candidate $f_0(1710)$ confirms BST’s 0^{++} at seat 11. Proton $m_p = C_2 \cdot \pi^5 \cdot m_e$ is the first (gap) rung of the same ladder; leptons (π^{-12} structure, no QCD content) are correctly OFF the gluonic ladder — the hadron/lepton split is content. So W2’s physical content is met on the interacting side; the result remains an honest conditional, now with the spectrum condition substantially DERIVED and only the rigorous interacting-RP upgrade formally residual. $\Delta = C_2 = 6$ SOLID; $C_2 = 2 \cdot N_c$ general-rank closed form (T2491). NOT ‘we proved YM’ — but the decisive probe fired and landed interacting.

2026-06-24 Wednesday (date-verified)

The Physical Yang–Mills Mass Gap via D_{IV}^5 Spectral Geometry

Paper A of the two-paper package (companion: Paper B, substrate uniqueness). v0.2.

Abstract

We study the Yang–Mills mass gap for the physical gauge group $SU(3)$ — color — derived (not assumed) from the substrate D_{IV}^5 in the companion Paper B. The mass gap value is the exact integer $\Delta = C_2 = 6$, the lowest nontrivial Casimir eigenvalue of the compact dual $Q^5 = SO(7)/[SO(5) \times SO(2)]$. The construction does not attempt to build 4D Yang–Mills from scratch on flat R^4 ; instead the D_{IV}^5 spectral theory directly supplies Osterwalder–Schrader reconstruction data (a rigorous Hilbert space, a gapped Hamiltonian, reflection positivity, 4D conformal/Poincaré covariance, clustering), so the existence of a rigorous gapped 4D theory follows from classical harmonic analysis on a bounded symmetric domain. The problem then reduces to a single question: is this gapped theory genuinely interacting Yang–Mills, or a (also-gapped, also-axiom-satisfying) generalized-free theory? In v0.2 we report that the decisive probe — the cross-channel glueball spectrum — has fired and landed on the interacting side: the four ground J^{PC} channels follow a linear conformal-energy ladder off one verified number (the genus of D_{IV}^5), reproducing the full spectrum in MeV at $<0.6\%$, with $2^{++} = g/n_C$ a blind prediction. The multiplicity itself is explained — the four channels are the irreducible components of $F \otimes F$, each forced by operator-algebra closure. The experimental scalar-glueball candidate $f_0(1710)$ confirms BST’s 0^{++} , and the proton sits on the same ladder as its first (gap) rung. We state the result honestly: the spectrum condition is now substantially derived; the lone formal residual is the rigorous interacting-RP upgrade. This is a decisive sharpening — not a claim that the Clay problem is solved.

1. Scope: the physical theory, not “all G ”

The Clay problem asks for a mass gap for *every* compact simple gauge group. BST addresses the physical theory: $G = SU(3)$, color. The companion Paper B proves D_{IV}^5 is the unique substrate and that $SU(3)$ is derived, not chosen (criteria-innocent uniqueness; $N_C = 3$ forced dimension-free). So “all G ” is not BST’s claim and not, on the BST view, a physical question: there is one substrate and it yields one gauge group. This paper makes claims only about the physical theory and is explicit about that scope (Section 9).

2. The mass gap value: $\Delta = C_2 = 6$ (SOLID)

The Yang–Mills Hamiltonian is identified with the D_{IV}^5 Casimir, whose spectrum on the compact dual Q^5 is the scalar-Laplacian tower $\{k(k+5)\}_{k \geq 0} = \{0, 6, 14, 24, \dots\}$. The lowest nontrivial eigenvalue — the gap — is

$$\Delta = C_2 = 6,$$

the $SO(7)$ vector Casimir, an exact integer from representation theory. Two structural facts sharpen it in v0.2:

- $C_2 = 2 \cdot N_C$ (general-rank closed form, T2491). The $(1,1)$ -type Casimir of $SO(n)$ is $2(n-2) = 2 \cdot N_C$; at the substrate’s rank it equals 6. (The rank-2 coincidences $C_2 = n_C + 1 = \text{rank}(\text{rank}+1)$ hold only here; the structural form is $C_2 = 2 \cdot N_C$.)
- The gap rung is physical. The proton is the first rung: $m_p = C_2 \cdot \pi^5 \cdot m_e = 938.25$ MeV. The 0^{++} glueball sits one genus higher on the absolute seat ladder, at seat $c_2 = C_2 + n_C = 11 \rightarrow 1720$ MeV.

Tier: SOLID (rep theory + the substrate Hamiltonian identification).

3. W4 dissolved: the gauge group is derived (Paper B)

A referee objection — “you treat only $SU(3)$ ” — is answered not by proving all G but by showing G is not a free input. Paper B establishes D_{IV}^5 as the unique irreducible Hermitian symmetric domain meeting prior, dimension-innocent criteria, forcing $\dim_C = 5$ and $N_c = 3$ (the multiplicity bound fixed by the rank via the proved Wallach Bottleneck Theorem T1829). So $SU(3)$ is substrate-derived. W4 (the “all G ” wall) is dissolved. v0.2 note: the whole dynamical content cascades from a single physical input — three colors fix rank = 2 ($N_c = \text{rank}^2 - 1$, T1829), and rank generates $\{n_C, C_2, g\}$ (T2491) — so the foundational claim tightens to “one physical input + the boundary integer $N_{\max}$.”

4. W1 folded: the D_{IV}^5 spectral theory supplies the OS data

The hard part of the Clay problem — *constructing* a rigorous interacting 4D QFT from scratch on R^4 — is not what BST does. The Osterwalder–Schrader reconstruction theorem builds a QFT from a data set, and the D_{IV}^5 spectral theory already carries that data:

OS datum	D_{IV}^5 realization	tier
rigorous Hilbert space	the Hardy space $H^2(D_{IV}^5)$	SOLID
Hamiltonian with a gap	the Casimir, gap = $C_2 = 6$	SOLID (Section 2)
reflection positivity	D_{IV}^5 tube-type (Cayley $\rightarrow T(\Omega)$ over the Lorentz cone); Euclidean-time reflection $\Theta = \text{cone}$ involution; OS-RP (free level) = Cauchy–Szegő reproducing-kernel positivity, automatic by RKHS	SOLID (free level)
4D conformal/Poincaré	$SO(4,2) \subset SO(5,2)$, UV-conformal/IR-gapped; Szegő-equivariance	SOLID-structural
clustering / unique vacuum	from the spectral gap $\Delta = C_2 = 6 > 0$ (exponential clustering)	SOLID

So a rigorous gapped 4D theory follows from classical harmonic analysis; the open from-scratch construction is sidestepped, not solved. At the free/spectral level reflection positivity, conformal covariance, clustering, and temperedness all hold; the single non-free-level piece is the interacting upgrade of reflection positivity — which is W2 itself. Tier: SOLID at free level; interacting-RP = W2.

5. W3 folded: net-compatibility via Bisognano–Wichmann (SOLID-CONDITIONAL)

The HS isometry intertwines the bulk and boundary operator nets, not merely the Hilbert spaces: HS is $SO(5,2)$ -equivariant (Hua–Korányi), and both the bulk Rehren net and the boundary net are modular reconstructions of the *same* $SO(5,2)$ positive-energy representation via Bisognano–Wichmann (Brunetti–Guido–Longo). So locality/causality transfer across HS. Tier: SOLID-CONDITIONAL on the BGL hypothesis-checks.

6. The one remaining question: interacting, or generalized-free?

After Sections 3–5 the prize is a single question: *is the OS-reconstructed gapped 4D theory genuinely interacting Yang–Mills, or a generalized-free gapped theory?* (A generalized-free field also satisfies the OS axioms and also has a gap.)

A trap we avoid. The boundary observables form a non-commutative Toeplitz algebra on H^2 (Hankel-compact commutators). “Non-commutative $\square$ interacting” is false — a free field also has a non-commutative CCR algebra. Interaction means nonzero connected n -point functions ($n \geq 3$).

The real locus of interaction. The Yang–Mills self-interaction is the vertex $[A,A]$, nonzero precisely because the gauge algebra is non-abelian. Therefore

interacting $\Rightarrow$ the gauge algebra is genuinely non-abelian $\mathfrak{su}(3) \Rightarrow$ the cross-channel glueball spectrum matches $SU(3)$ (W2).

Three faces of one gauge-structure question, all in the $\mathfrak{so}(7) \supset \mathfrak{g}_2 \supset \mathfrak{su}(3)$ of Section 8. Two pieces of evidence toward the interacting side, both strengthened in v0.2:

- The bulk-color algebra closes as non-abelian $\mathfrak{su}(3)$. The bilinear (Schwinger) realization closes into $\mathfrak{su}(3)$ — CCR + all 81 $\mathfrak{gl}(3)$ commutators + the A_2 relations (Elie toy 4301). The substrate identification (that the bulk-color Toeplitz operators *are* these bilinears) is framework, in progress.
- The spectrum is non-additive and now derived (Section 7). A generalized-free theory has an additive multi-particle spectrum and no genuine bound-state $J^{\wedge}PC$ tower. The derived ladder (Section 7) is non-additive and channel-structured — the fingerprint of binding. In v0.2 this is no longer a caveated lead but a derived spectrum at $<0.6\%$.

7. The decisive probe: the cross-channel glueball spectrum (W2) — DERIVED (the v0.2 upgrade)

In v0.1 this section was *named-open at structural tier*. It is now derived. The probe fired, and it lands on the interacting side.

7.1 Why there are exactly four channels (operator-algebraic closure — the WHY)

The gauge field F is a rank-2 antisymmetric tensor operator on $H^2(D_{IV}^5)$. Its bilinears $F \otimes F$ decompose into exactly four irreducible $J^{\wedge}PC$ components, and each must have eigenstates for the substrate’s operator algebra to close:

channel	operator	what it commits
0^{++}	$\text{Tr}(F^2)$	energy density (the action)
0^{-+}	$\text{Tr}(F \tilde{F})$	topology (Pontryagin density)
2^{++}	$\text{Tr}(F^{\wedge\mu\rho} F^{\wedge\nu}_{\rho})$ traceless	curvature (the stress tensor; couples to gravity)
1^{+-}	$\text{Tr}(F[D,F])$	derivative structure

If 0^{-+} did not exist, topology could not be committed; if 2^{++} did not exist, curvature could not couple to gravity. The multiplicity is forced by the tensor structure, not chosen — the answer to the reviewer’s “why four channels rather than one or twelve?” This is the same architectural principle as nuclear magic numbers, with a different driver per operator class: Pauli antisymmetry for fermion content (shells $\rightarrow 28, 50, 82, 126$), operator-algebra closure for the bosonic gauge tensor (channels $\rightarrow J^{\wedge}PC$ spectrum). The oddballs ($0^{+-}, 1^{-+}, 2^{+-}$) are forbidden at two gluons — clean, unmixable experimental channels.

7.2 The masses: one verified number sets the whole ladder

The glueball mass is the eigenvalue of the linear conformal Hamiltonian (the $SO(2)$ dilatation generator) on the holomorphic discrete series on $H^2(D_{IV}^5)$ — diagonal in the K-type basis by Schur:

$$m \propto E = \lambda_0 + (\text{energy step}), \lambda_0 = \text{genus}(D_{IV}^5) = n_C = 5.$$

The genus is verified from the multiplicity formula $p = (r-1)a + b + 2 = 5$. The energy step is the $SO(5)$ harmonic degree = spin J ; the parity-odd (Hodge-dual) sector carries the half-canonical twist $n_C/2$ (the canonical bundle transforms with the genus). With one dimensionful anchor (seat = $\pi^5 \cdot m_e = 156.4$ MeV; $m(0^{++}) = c_2 \cdot \text{seat} = 1720$ MeV) the ratios give absolute masses:

channel	step	ratio	substrate form	BST (MeV)	lattice (MeV)	dev
0^{++}	0	1	—	1720	1730	−0.6%
2^{++}	$J = 2$	$7/5$	g/n_C (blind)	2408	2400	+0.3%
0^{-+}	$n_C/2$	$3/2$	N_c/rank	2580	2590	−0.4%
1^{+-}	$1 + n_C/2$	$17/10$	—	2924	2940	−0.5%

The $2^{++} = g/n_C$ leg is blind — genus + spin only, nothing read from the data — and it lands at 0.3% because $g = n_C + \text{rank}$. The four ratios collapse to two BST-natural numbers (the genus n_C and the half-canonical $n_C/2$), not four free ratios: that $2 \rightarrow 4$ over-constraint is what distinguishes a derivation from a coincidental match. (Quantization note: the channel *ratios* live on the rung ladder $\lambda_0 = n_C$; the absolute scale comes only from anchoring 0^{++} at seat 11. These are two distinct quantizations and only 0^{++} sits at an integer seat.)

7.3 The radial direction is linear, by Schur (a theorem, with a clean negative)

The scalar holomorphic discrete series is one irreducible representation; by Schur’s lemma the Casimir is constant across all its K-types, so it cannot distinguish radial levels. The first radial scalar excitation $0^{++*} =$ the (1,1) r^2 -mode therefore sits at the linear conformal energy $E = \lambda_0 + 2 = 7$, degenerate with 2^{++} (2408 MeV; lattice ~2670, within quenched excited-gluon error). This retires the mass-linear-vs-mass²-quadratic ambiguity: within an irrep the spectrum is linear; any quadratic structure is strictly *inter*-irrep (the discrete-series tower {0,6,10,12,14,...} of Grace’s T2490). A consequence stated honestly: the appealing “spin-linear / radial-quadratic factorization” (a Curvature-Principle-at-operator-level reading) is a clean negative — both directions are linear within the irrep.

7.4 Experimental confirmation and the decay-side picture

The experimental scalar-gluon candidate $f_0(1710) \approx 1704$ MeV matches BST’s 0^{++} at seat 11 (1720 MeV) — the right state confirming the lightest predicted gluon. The 0^{-+} topological coupling is substrate-computable: f_G is the 0^{-+} Bergman mode-norm, a Gindikin-Gamma ratio at BST arguments (the 0^{++} kernel = $60 = C_2 \cdot n_C \cdot \text{rank}$ exactly); with $m(0^{-+}) = 2580$ MeV and the established pure-gluon $\chi_{\text{top}}^{1/4} = 180$ MeV the Witten-Veneziano identity $\chi_{\text{top}} = f_G^2 \cdot m^2(0^{-+})$ gives $f_G(0^{-+}) = 12.6$ MeV (the absolute χ_{top} prediction is one volume normalization away). The decay-side mechanism is mixing (gluon $\leftrightarrow$ $q\bar{q}$ overlap on the shared Hardy space, matching $f_0(1500)/f_0(1710)$ phenomenology), not a “dump.”

7.5 Honest disposition of §7

The cross-channel spectrum — named-open in v0.1 — is derived: four channels at <0.6%, one blind leg ($2^{++} = g/n_C$), multiplicity explained by operator-algebra closure, radial direction settled by Schur, the lightest state experimentally confirmed ($f_0(1710)$). A generalized-free theory produces none of this — no bound-state J^{PC} tower, no non-additive ladder. So the physical content of W2 lands on the interacting side. What remains is the *formal* step (Section 9): upgrading this physical evidence to the rigorous interacting-RP statement.

8. The $so(7)$ unification (LEAD-STRENGTHENED)

Color and the Yang–Mills spectrum live in one algebra: $su(3) \subset g_2 \subset so(7)$, where g_2 is the octonion 3-form stabilizer ($14 = 8 \oplus 3 \oplus \bar{3}$; the $7 = 3 \oplus \bar{3} \oplus 1$), and $so(7)$ is the compact dual isometry of Q^5 — the same $so(7)$ whose Casimir spectrum is the gluon tower of Section 2. So the color group and the gauge-boson spectrum are unified in one $so(7)$; $g = 7$ is the $so(7)$ vector = g_2 fundamental. Color is not a geometric isometry of the noncompact domain ($su(3) \not\subset so(5,2)$ compactly) but is a geometric subalgebra of the compact-dual $so(7)$; on the domain side it is the operator algebra on H^2 . v0.2 note: the discrete-series spectrum of this $so(7)$ has the substrate primaries $\{N_c, n_C, C_2, g\}$ as its lowest half-Casimirs (T2490) — the gap $C_2 = 6$ is the first rung, and the gluon ladder is its linear-energy face. Tier: LEAD-STRENGTHENED — the algebraic picture is whole; the bilinear-Schwinger operator realization on H^2 is in progress.

9. Honest scope: the conditional, materially advanced

We do not claim to have proved the Yang–Mills mass gap. The honest statement, updated for v0.2:

Given (i) the OS bounded-checks (reflection positivity, locality via Section 5) hold to executed rigor including the interacting-RP upgrade, the physical SU(3) Yang–Mills theory exists as a rigorous 4D QFT with mass gap $\Delta = C_2 = 6$ — and (ii) the cross-channel spectrum, which in v0.1 was the open identifier of interacting-vs-free, is now DERIVED (Section 7), landing the physical content on the interacting side.

What is SOLID independent of the conditions: the gap *value* $C_2 = 6$ (Section 2); the substrate-derivation of SU(3) (Paper B); the OS *data* (Section 4); the so(7) unification (Section 8); and now the cross-channel spectrum (Section 7, four channels $<0.6\%$, blind 2^{++} leg, $f_0(1710)$ confirmation). What remains: the formal interacting-RP upgrade — turning the derived physical spectrum into the rigorous OS interacting statement. v0.1 was a two-condition conditional with both open; v0.2 reports condition (ii) substantially met and isolates the single formal residual. This is a sharpening of the Clay problem to one decidable formal step, with the physics evidence already on the interacting side — not a claim of its solution.

10. Open items and falsifiers

- Formal interacting-RP upgrade (Section 9): the lone residual — promote the derived physical spectrum to the rigorous OS interacting statement.
- #418 substrate identification: that the bulk-color Toeplitz octet *is* the bilinear-Schwinger su(3) on H^2 (algebra closure verified; identification in progress).
- Blind-leg audit promotion: confirm $\lambda_0 = \text{genus} = n_C$ is the Bergman lowest weight, and canonical-bundle weight = genus (Elie toys A+B) — promotes 2^{++} from blind-at-lattice to blind-full-audit and the rest of the spectrum from genus alone.
- Falsifier (sharpened, not retired): the spectrum is a forward prediction — 0^{+++} degenerate with 2^{++} at 2408 MeV is testable by precise lattice spectroscopy; if 0^{+++} is cleanly above 2^{++} , it must be a higher irrep (its J^{PC} forced, not fit). If the cross-channel ratios were to fail, the interacting reading fails (the gap value $C_2 = 6$ and Paper B survive).

Count HOLDS 4 of 26. SU(3) scope; glueball masses are predictions, not SM parameter reductions. Paper A v0.2: the wall map collapsed to one question, and the decisive probe (the cross-channel spectrum) has now fired and landed interacting — four channels at $<0.6\%$, a blind $2^{++} = g/n_C$ leg, multiplicity explained by operator-algebra closure, the lightest state confirmed by $f_0(1710)$, the proton on the same ladder as its gap rung. The gap value is SOLID; the result is an honest conditional with the spectrum condition substantially derived and one formal residual. INTERNAL.

Draft v0.2. Spine: Grace's OS reduction (W1) + net-compat (W3) + T2490/T2491 spectral cascade; Lyra's W1=#418=W2 reduction + the linear-conformal-Hamiltonian glueball ladder + Schur radial settlement; Elie's Schwinger closure (4301) + proton-gap-rung; the so(7) unification; Paper B (W4); operator-algebraic-closure multiplicity (K498, Casey Wed AM). Companion to Paper B v0.5.

— Lyra, Wed 2026-06-24 (date-verified). Paper A v0.2, §7 spectrum derived.